

Chapter 10: Reflection of Light - Summary

1. Definitions

- **Reflection:** The "bouncing back" of light when it hits a boundary.
- **Normal:** An imaginary line perpendicular (90°) to the surface.
- **Incident Ray:** The incoming light beam.
- **Reflected Ray:** The outgoing light beam.

2. The Laws of Reflection

1. The incident ray, reflected ray, and normal all lie in the **same plane**.
2. **Angle of Incidence = Angle of Reflection** ($\hat{i} = \hat{r}$). *Note: Always measure angles from the Normal, not the surface!*

3. Types of Reflection

Regular (Specular)	Diffuse
Smooth surfaces (mirrors)	Rough surfaces (paper, walls)
Parallel rays stay parallel	Rays scatter in many directions
Forms a clear image	Allows us to see the object itself

4. Plane Mirror Images

Images in a flat mirror are:

1. **Virtual** (Behind the mirror, cannot be projected).
2. **Upright** (Same way up).
3. **Same Size** as the object.
4. **Same Distance** (Object distance p = Image distance q).
5. **Laterally Inverted** (Left becomes right).

5. Field of Vision

The area you can see in a mirror. You can see more by:

- Getting closer to the mirror.
- Using a larger mirror.
- Using a **Convex Mirror** (curves outward).